

SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26200
- **Problem Statement Title**- Student Innovation- Innovative Ideas that help manage and generate renewable/ sustainable sources more efficiently
- **Theme**- Renewable / Sustainable Energy
- **PS Category**- Software
- **Team ID**- SIH26-A0H-T139
- **Team Name** – ELECTRASIX



Challenge We're Solving

- **Solar Goes Unused:** surplus is exported or wasted at midday, while flexible loads still run at night on grid power — apps only report numbers, never what to run.

Our Solution

- **Live Solar Opportunity Engine:** Matches real-time surplus generation against your registered appliances and tells you exactly what to run right now.
- **Solar Queue Scheduler:** Flexible loads like pumps and EV chargers queue automatically and trigger the moment enough surplus appears.
- **Priority-Ladder Dashboard:** One home screen shows generation, direct use, storage and surplus — no electrical-engineering knowledge required.

Addressing the Core Problem

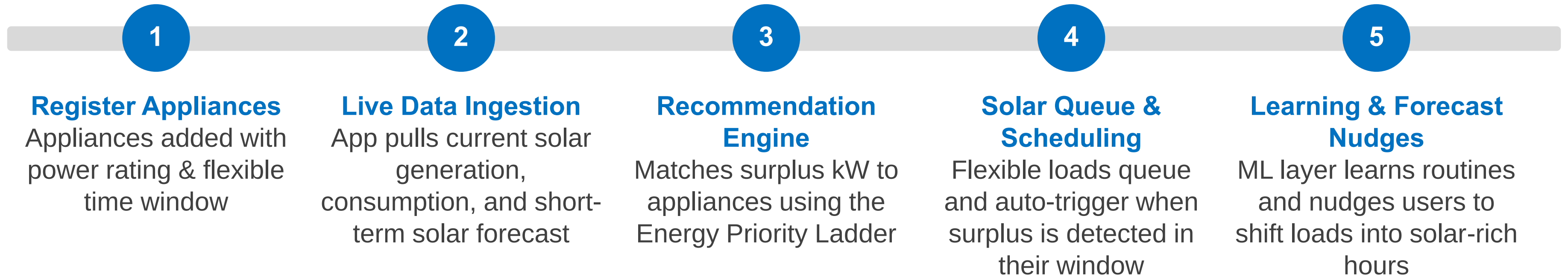
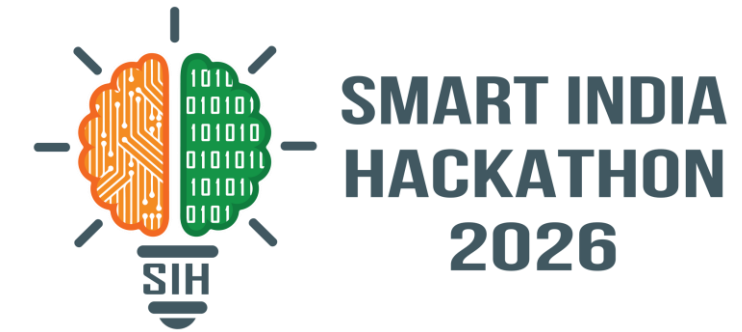
1. **Zero Extra Hardware:** Works with existing inverter/meter data and appliance profiles — no smart plugs required for the MVP.
2. **Zero Guesswork Gap:** Real-time "Run This Now" alerts turn a confusing energy dashboard into one clear action.
3. **Nothing Wasted:** An Energy Priority Ladder ranks essential, flexible, battery, EV and pump loads before any export.
4. **Scales Anywhere:** Same engine works for a single home, a hostel block, or a full campus microgrid.

What Makes Us Different

1. **Forecast-Driven, Not Reactive:** Combines live generation with a solar forecast to recommend action before surplus even happens.
2. **Built-In Solar Queue:** Washing machines, pumps and EV chargers auto-queue into high-solar windows instead of running at night.
3. **Learns the Routine:** An AI layer learns usage patterns and suggests shifting habitual loads into peak-solar hours.
4. **Priority-Ladder Logic:** Essential → Flexible → Battery → EV → Pump/Heater → Export — curtailment is the last resort.
5. **Free-Tier Feasible:** Built on open forecasting APIs and free-tier infrastructure, keeping deployment cost near zero.

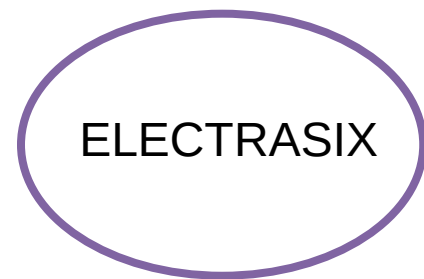


TECHNICAL APPROACH

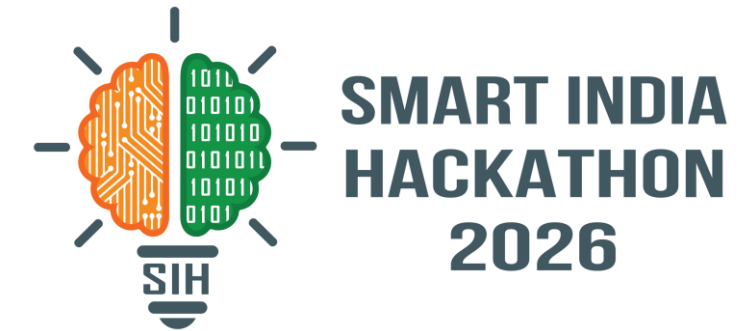


Technologies Used





FEASIBILITY AND VIABILITY



Challenges

Solutions

Feasibility Categories

Technical

- Built entirely on free-tier tools — open weather APIs, Supabase, Vercel/Railway; forecasting runs as a lightweight service.

Cost

- No paid APIs or GPU servers — near-zero cost even at full Campus/ microgrid scale.

Delivery

- Fully buildable within the hackathon timeframe using open-source forecasting libraries.

Forecast Reliability

- Solar forecast accuracy varies with local cloud cover; inverter data formats differ across manufacturers and models.

Load-Matching Accuracy

- User-entered appliance power ratings may be inaccurate, causing mismatched run recommendations.

Adoption & Automation Trust

- Users may distrust auto-scheduling of flexible loads, or ignore recommendation notifications entirely.

Multi-Source Forecast Blending

- Blend multiple weather sources with rolling-error correction; a generic adapter layer normalizes common inverter formats.

Default Ratings + Calibration

- A default power-rating library per appliance type, refined by a one-time calibration against actual draw.

Recommend-First, Opt-In Automation

- Starts in recommend-only mode with manual approval; auto-trigger via smart plug unlocks only after opt-in.

Up to 100% Self-Consumption • 6+ Appliance Categories • Zero Extra Hardware for MVP

Target Audience Impact

Homeowners

- Solar Queue shifts flexible loads into daylight hours, cutting grid electricity bills.

Campuses & Institutions

- Priority-ladder scheduling across hostel blocks and departments maximizes shared solar use.

Utilities & Grid

- Less uncoordinated export smooths local feeder-level demand swings.

Benefit Analysis

Economic

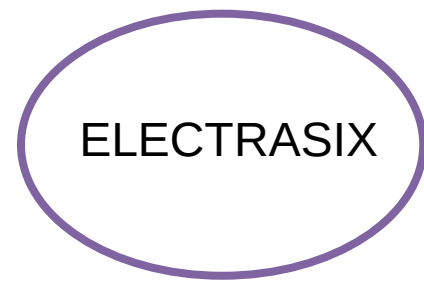
- Lower electricity bills by shifting consumption to self-generated solar instead of costly grid import.

Environmental

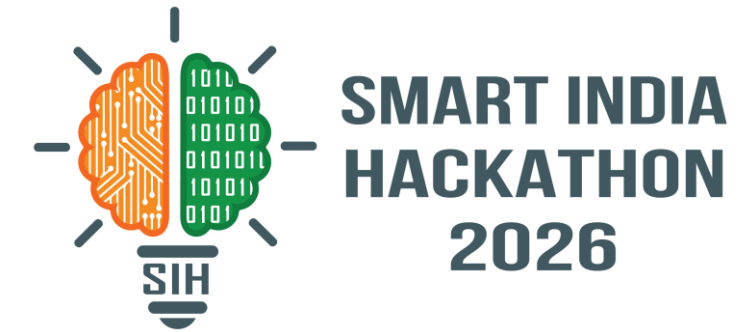
- Less curtailed or wasted renewable energy and lower reliance on carbon-intensive peak-hour grid power.

Scalability

- Same engine scales from one household to a multi-building campus microgrid with no architecture change.



RESEARCH AND REFERENCES



Solar & Forecasting

- NREL PVWatts (Solar Forecasting) — pvwatts.nrel.gov — reference methodology for photovoltaic generation estimation

Standards & Data

- Open-Meteo Weather API — open-meteo.com — free-tier weather & solar irradiance data used for forecasting

Government Alignment

- National Solar Mission (MNRE) — mnre.gov.in — Government of India initiative on solar adoption that this solution directly supports
- PM Surya Ghar: Muft Bijli Yojana — pmsuryaghar.gov.in — rooftop solar scheme relevant to the target user base

Backend Infrastructure

- Supabase Documentation — supabase.com/docs — Postgres + Auth + Realtime backend used for scheduling and user data